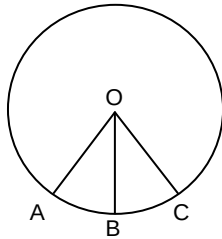


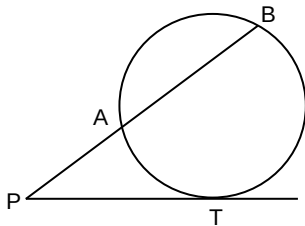
GEOMETRY_HANDOUT 02

Directions for questions 1 to 15: Solve the following questions.

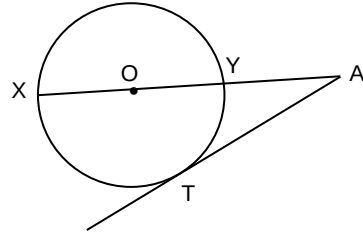
1. In trapezium PQRS, $PQ \parallel RS$. If $PQ = 10$, $QR = 17$, $RS = 31$ and $PS = 10$, find the height of the trapezium.
2. ABCD is a parallelogram. If $AB = 6$ cm and $BC = AC = 8$ cm, find BD.
3. The greatest distance between any two points on a rhombus is $\sqrt{3}$ times the length of the shorter diagonal, which is 2 cm. Find the side of the rhombus.
4. Find the number of diagonals in a polygon with 20 sides.
5. The number of diagonals in a regular polygon is 35. Find the measure of each exterior angle of the polygon.
6. The measure of each interior angle of a regular pentagon is thrice the measure of each exterior angle of another regular polygon (P). Find the measure of each interior angle of P.
7. ABCDEF is a regular hexagon with side 10 cm. Find the length of the diagonal BD.
8. In the given circle, O is the centre and $\angle AOB = \angle BOC = 50^\circ$. Find $\angle ABC$.



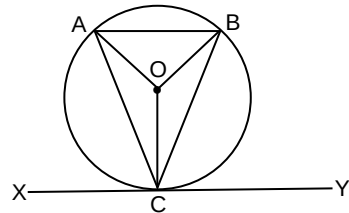
9. The distance between two equal and parallel chords in a circle is 32 cm. If the radius of the circle is 20 cm, find the length of each chord.
10. In the figure, PT is a tangent to the circle and PAB is a secant. If $PA = 8$ cm and $PT = 12$ cm, find AB.



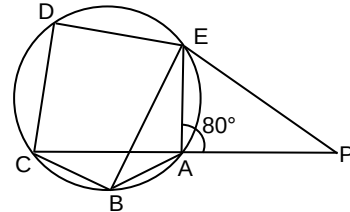
11. In the figure, AT is a tangent to the circle centered at O. The radius of the circle is 10 cm. If $\angle TOY = 60^\circ$, find the length of AX.



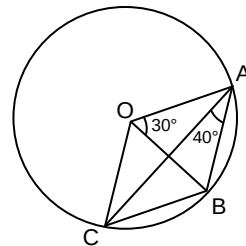
12. In the figure, O is the centre of the circle and XY is a tangent to the circle at C. If $\angle BCY = 70^\circ$, find $\angle BOC$.



13. In the figure, PC is a secant and $\angle PAE = 80^\circ$. Find $\angle CBE$.



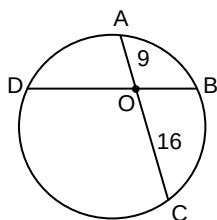
14. Find the length of a direct common tangent drawn to two circles whose centres are 20 cm apart and whose diameters are 30 cm and 6 cm.
15. In the figure, O is the centre of the circle. If $\angle AOB = 30^\circ$ and $\angle CAB = 40^\circ$, find $\angle ABC$.



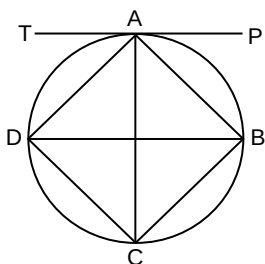
ADDITIONAL QUESTIONS FOR PRACTICE

Directions for questions 1 to 5: Solve the following questions.

1. In the figure, O is the point of intersection of the chords AC and BD. Find the minimum length of BD.



2. ABCD is a cyclic quadrilateral and PAT is a tangent. If $\angle PAC = 70^\circ$, find $\angle ABC$.



3. In cyclic quadrilateral PQRS, the ratio of the measures of the external angles at P and R is 2 : 3. Find $\angle QRS$.
4. The number of diagonals of a polygon is numerically equal to 3 percent of the sum of the measures of all the internal angles (in degrees) of the polygon. Find the number of sides of the polygon.
5. Two parallel chords are drawn on the opposite sides of the centre of the circle. The radius of the circle is 30 cm and the lengths of the chords are 48 cm and 36 cm. Find the distance between the two chords.